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13. ABSTRACT

Describes a method for evaluation of paving equipment operational and functional performance characteristics. Identifies supporting test, facilities, and equipment required. Specifies procedures for operator training, photographic coverage, safety, initial inspection, physical characteristics, human factors, electromagnetic interference, performance, environmental effects, maintenance, reliability, transportability, durability, and value analysis.

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14. KEY WORDS	LINK A		LINK B		LINK C	
	ROLE	WT	ROLE	WT	ROLE	WT
Aggregate Spreaders & Driers						
Asphalt Paving Equipment						
Concrete Preparation Equipment						
Crushing Plant						
Curing Machine						
Paving Equipment						
Rollers						
Sweeper (Road)						

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REFERENCE FILE

U.S. ARMY TEST AND EVALUATION COMMAND
SYSTEM ENGINEERING TEST OPERATIONS PROCEDURES

AMSTE-RP-702-108
Test Operations Procedure 9-2-111 12 April 1972

PAVING EQUIPMENT

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SECTION I
GENERAL

1. Purpose and Scope. This TOP describes test procedures for evaluating the operational and performance characteristics of paving equipment. Equipment covered includes: soil sampling kit; road rooter, roller and sweeper; batching plant; drier; mixer; asphalt melter; paving, spreading, finishing and curing machine. From the tests listed in Section II, the test director can select those that will satisfy the requirements for the particular test item and the particular test type (i.e., engineering test, initial production test, etc.). Test objectives are to determine conformance of the test items to QMR, MN or other suitability criteria. For initial production tests, scope will be in accordance with the contractual provisions of the applicable military specifications and suitability criteria established by the test directive. Environmental tests, as dictated by the size and nature of the test item, may require use of components or sample sections of materials as well as on-site climatic tests.

2. Background. Military operations involving the movement of aircraft, vehicles, and personnel increase in efficiency when paved surfaces are available for use. Construction of temporary and permanent roads, ramps, and runways is therefore culminated, when possible, with the application of asphalt or concrete surfaces. The paving operation normally involves three phases of action, namely: road bed preparation, material preparation-processing, and paving. The road bed preparation

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phase consists of probing the area for the proposed road to determine suitability for construction of the road and the preparation of the road bed using earth moving and loading equipment including specialized equipment such as rooters, rollers and sweepers; material preparation-processing requires the use of batching plants, drier-mixers for bituminous and concrete, and asphalt melters; the paving phase uses bituminous paving machines and concrete spreaders, finishing and curing machines.

3. Equipment and Facilities. Equipment and facilities required are defined in the documents listed in Section II.

SECTION II
TEST PROCEDURES

4. Supporting Tests. Subtests (generally in preferred order of completion with respect to high-risk, short duration) to be considered in formulating a test plan are listed below with references.

TEST SUBJECT TITLE	PUBLICATION NO.
a. Preoperational Inspection	2-2-502
(1) Operator Training and Familiarization	10-2-501
(2) Photographic Coverage	7-3-519
(3) Preoperational Checks (refer to para 5)	
b. Physical Characteristics	10-2-500
(1) Magnetic Particle Test	MIL-STD-271D para 4
(2) Liquid Penetrant	para 5
(3) Hardness Test	ASTM E18
c. Safety (refer to para 6)	10-2-508 2-2-508
d. Compatibility with Related Equipment	2-3-512
e. Performance Tests	
(1) Road Bed Preparation Equipment	
(a) Soil Sampling Kit	MIL-S-51210 para 4.6
(b) Road Rooter	MIL-R-587A para 4.2.1, 4.2.2

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TEST SUBJECT TITLE	PUBLICATION NO.
(c) Roller	MIL-R-10550E, para 4.6.2 MIL-R-52499A MIL-R-52586, para 4.6.2
(d) Sweeper (Towed and Self-Propelled)	MIL-S-18141D, para 4.3.2 4.3.3, 4.3.5, 4.3.6 MIL-S-27153D, para 4.6
(2) Material Preparation - Processing Equipment	
(a) Batching Plant (Fixed and Wheel-Mounted)	MIL-B-22353A, para 4.3 MIL-B-3755A, para 4.3.2
(b) Driers - Mixers (Bituminous - Concrete)	MIL-D-3968B, para 4.6.2
(c) Asphalt Melter	MIL-M-52182C, para 4.6.2
(3) Paving Equipment	
(a) Bituminous Paving Machine	MIL-P-335E, para 4.6.2
(b) Concrete Spreader	MIL-S-16328C, para 4.6.2
(c) Concrete Finishing Machine	MIL-F-82098A, para 4.3
(d) Concrete Curing Machine	MIL-C-52617, para 4.6.2
f. Environmental Tests	
(1) Temperature	AR 70-38
(2) Rain	2-2-815
(3) Fungus	4-2-818
(4) Salt Fog	MIL-STD 810B, Method 509
(5) Electromagnetic Interference Characteristics	MIL-STD-461A, Notice 4 MIL-STD-462, Notice 3, Method RE05
g. Transportability	1-2-500 MIL-STD-209C

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TEST SUBJECT TITLE	PUBLICATION NO.
h. Human Factors Evaluation	2-2-803 10-2-505
Noise Levels	3-2-811
i. Reliability	AMCP 702-3
Confidence Intervals Sampling Size	3-1-002
j. Durability (Endurance Testing)	2-2-507 10-2-502
k. Maintenance Evaluation	1-2-507 10-2-507 TECR 750-15
l. Value Analysis	AMC SUPPL 1 to AR 11-26

SECTION III
SUPPLEMENTARY INSTRUCTIONS

5. Preoperational Checks.

a. Objective. To determine that the test item is ready for operation.

b. Method. The test item is inspected to insure that all types of preservation have been removed and all accessories removed for transporting test item have been securely attached or installed. A check is made to determine that all lubricants, hydraulic fluids, coolants and battery electrolytes used conform to the specifications and required operating levels indicated on charts or tags provided with the test item. All controls and indicators are exercised to insure proper operation and accuracy of indications.

c. Data Required.

(1) Nomenclature and type of test item.

(2) Specified and actual lubricants, hydraulic fluids, coolants and battery electrolyte types and operating levels.

(3) Ease of movement of operating controls.

(4) Accuracy of indicator displays.

(5) Any evidence of defects encountered.

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d. Analytical Plan. The recorded data are analyzed and compared with the requirements of the MN to determine conformance to specifications.

6. Safety. The applicable QMR, MN or specification is carefully reviewed for safety criteria. Certification of safety aspects determined during construction or preproduction tests are obtained from the developer and accepted where valid. All personnel, including operators, are made thoroughly familiar with all safety procedures and requirements of the test item prior to test operations. Maximum precautions are taken during stability, overload and other safety critical phases. Proper performance of the test item during these operations is assured by using experienced personnel during pretest operations. In addition to MTPs 2-2-508 and 10-2-508 and test item technical manuals, documents such as the following are reviewed for applicability:

AMC Safety Manual 385-100.

MIL-STD-882, "Systems Safety Program for Systems and Associated Subsystems and Equipment."

National Safety Council Data Sheet No. 130, "Safe Use of Heavy Duty Equipment on Construction Jobs."

Underwriters' Laboratories, UL 558, "Standards for Safety for Power Operated Industrial Trucks."

American Standards Association Bulletin 56.1, "Safety Code for Industrial Trucks."

7. Salt Fog Test. (Climatic Test Site)

a. Objective. To determine the degree of corrosion the test item can withstand without degradation of performance.

b. Method. The test item is operated in shallow water, near shore, exposed to salt water and salt spray for the period of time stated in the MN. After exposure, the test item is statically stored without washing. At the end of 120 hours of storage, the test item is inspected for corrosion and again operated exposed to salt water and salt spray. The test item is then statically stored, without washing, for another 120 hours. At the conclusion of the second storage period, the test item is inspected for corrosion and operated to determine any degradation to performance.

c. Data Required.

(1) Nomenclature and type of test item.

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(2) Location of test site.

(3) Periods of time test item operated exposed to salt water and salt spray.

(4) Actual storage times.

(5) Inspection results first and second inspections.

(6) Degradation to performance due to corrosion.

d. Analytical Plan. The recorded data are summarized and analyzed and the degree of degradation to performance determined. This evaluation is compared with the requirements of the MN to determine conformance to specifications.

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APPENDIX
REFERENCES

1. AR 70-38, "Research, Development, Test and Evaluation of Materiel for Extreme Climatic Conditions."

2. AMC Supplement 1 to AR 11-26, "Value Engineering."

3. AMCP 702-3, "Reliability Handbook."

4. TECR 750-15, "Maintenance Evaluation during Testing."

5. MIL-STD-209C, "Slinging Eyes and Attachments for Lifting and Tying Down Heavy Military Equipment."

6. MIL-STD-271D, "Nondestructive Testing Requirements for Metals."

7. MIL-STD-461A, "Electromagnetic Interference Characteristics, Requirements for Equipment", including notices 1 thru 4.

8. MIL-STD-462, "Electromagnetic Interference Characteristics, Measurement of", including notices 1 thru 3.

9. MIL-STD-810B, "Environmental Test Methods", including notices 1 thru 4.

10. MIL-B-3755A, "Batching Plant, Cement, 50-Ton-Per-Hour Capacity."

11. MIL-B-22353A, "Batching Plant, Aggregate, (Wheel-Mounted, 30-Ton Capacity)."

12. MIL-C-52617, "Curing Machine, Concrete, Wheel-Mounted."

13. MIL-D-3968B, "Drier-Mixers, Bituminous-Concrete Materials, Wheel-Mounted."

14. MIL-F-82098A, "Finishing Machines, Concrete-Paving, Slip-Form, Full-Tracked."

15. MIL-M-52182C, "Melter, Asphalt", including amendment 1.

16. MIL-P-335E, "Paving Machines, Bituminous Material: Crawler-Mounted, Gasoline-or Diesel-Engine-Driven."

17. MIL-R-587A, "Rooter, Road, Cable-Controlled."

18. MIL-R-10550E, "Roller, Motorized, Tandem, 2-Axle and 3-Axle, Ged or Ded."

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19. MIL-R-52499A, "Rollers, Towed, Smooth Drum, Vibratory, Airmobile, Single-Drum."

20. MIL-R-52586, "Rollers, Towed, Sheepsfoot, High Speed."

21. MIL-S-51210, "Soil Sampling Kit."

22. MIL-S-16328C, "Spreader, Concrete, Form Riding, 20-To-25-Foot Width."

23. MIL-S-18141D, "Sweeper, Rotary, Towed."

24. MIL-S-27153D, "Sweeper, Rotary, Self-Propelled A/S32U-13."

25. American Society for Testing and Materials (ASTM), "E18-Rockwell Hardness Test."

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